

Odong Stephen

Biomedical Engineer

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Highly motivated Biomedical Engineer with 4 years of experience in analyzing complex medical and biological problems, coupled with strong critical thinking and communication abilities. Demonstrated expertise in biomedical equipment installation, precise adjustment, and rigorous safety evaluation, ensuring optimal performance and patient well-being. Certified in Artificial Intelligence (AI) Fluency and AI Agents Fundamentals, proficient in Python, SolidWorks, and Unity for innovative solution development, including virtual reality tools for engineering education and advanced diagnostics. Possesses a proven track record in project management, technical support, and contributing to impactful research initiatives.

WORK EXPERIENCE

Research Assistant, Virtual Reality Lab

Mbarara University | Mbarara, Uganda | 2025-05 – Present

- Led the development and implementation of an innovative virtual reality (VR) tool for biomedical engineering education using Unity 3D and C#, enhancing practical learning experiences for over 100 students.
- Designed and integrated interactive 3D anatomical models with SolidWorks and Blender, significantly improving visualization and comprehension of complex biological systems.
- Collaborated closely with faculty and researchers to define project scopes and gather user requirements, ensuring alignment with curriculum objectives and delivering high-fidelity simulations.
- Conducted rigorous user testing and iterative design cycles, leveraging feedback to optimize VR tool functionality and user engagement, resulting in a 20% increase in student retention of key concepts.

Biomedical Engineer Intern

Train Biomedics Initiative Uganda | Uganda | 2026-06 – 2026-08

- Performed comprehensive installation and preventative maintenance on over 50 critical biomedical devices, including Anesthesia machines, suction machines, and laboratory analyzers, ensuring peak operational efficiency.
- Conducted detailed safety evaluations and risk assessments for medical equipment, adhering to international standards and reducing potential hazards by 15% across various clinical settings.
- Provided prompt and effective technical support and troubleshooting for medical staff, resolving equipment malfunctions efficiently and minimizing downtime for patient care facilities.
- Assisted in the development and delivery of targeted training programs for junior technicians and end-users, enhancing their proficiency in operating and basic maintenance of sophisticated medical technology.
- Documented all service activities and equipment histories in a centralized management system, contributing to a 10% improvement in inventory and maintenance tracking accuracy.

Biomedical Intern

Train Biomedics Initiative Uganda | Uganda | 2025-06 – 2025-07

- Supported senior biomedical engineers in routine inspections and functional testing of diverse medical equipment, ensuring compliance with operational guidelines and manufacturer specifications.
- Gained hands-on experience in disassembling, cleaning, and reassembling complex medical devices, enhancing understanding of internal components and systems.
- Participated actively in inventory management and procurement processes for spare parts and consumables, optimizing resource allocation for equipment maintenance.
- Shadowed diagnostic procedures and patient monitoring activities, developing a practical understanding of equipment application and workflow in clinical settings.
- Contributed to data collection and report generation for equipment performance metrics, aiding in long-term maintenance planning and predictive analytics.

Biomedical Engineer Volunteer

Holistic Rehabilitation Center Uganda | Uganda | 2024-11 – 2024-12

- Provided essential technical support and maintenance for critical rehabilitation equipment, including therapeutic exercise machines and mobility aids, significantly improving patient access to vital tools.
- Assisted in customizing and adapting assistive devices for individuals with specific needs, directly enhancing patient mobility and overall quality of life.
- Collaborated effectively with medical staff to ensure safe and effective use of biomedical equipment, directly contributing to the center's patient safety protocols.
- Documented equipment status and service records comprehensively, ensuring accurate tracking of assets and maintenance history for regulatory compliance and future planning.

EDUCATION

Mbarara University of Science and Technology

Bachelor of Biomedical Engineering
2023-09 – 2027-10

Amazing Grace High School

Uganda Advanced Certificate of Education (UACE)
2021-01 – 2022-12

Kitgum High School

Uganda Certificate of Education (UCE)
2016-02 – 2019-11

Kitgum P.7 School

Primary Leaving Examination
2008-02 – 2015-11

SKILLS

Biomedical Engineering, Medical Device Maintenance, Equipment Troubleshooting, Clinical Research, Regulatory Compliance, Diagnostic Equipment, Therapeutic Devices, Patient Safety, Artificial Intelligence (AI), Machine Learning, Python Programming, C#, Virtual Reality (VR) Development, Unity 3D, SolidWorks, 3D Design, Data Analysis, Project Management, Leadership, Technical Documentation, Microsoft Office Suite, Problem Solving, Communication, Teamwork and Collaboration, Time Management, Web Development, Risk Assessment

PROJECTS

Virtual Reality Tool for Biomedical Engineering Education

<https://stephstar.github.io/portfolio>

(Unity 3D, C#, SolidWorks, Blender, Virtual Reality (VR), User Interface (UI) Design)

Designed and developed an interactive virtual reality application using Unity 3D and C# to significantly enhance practical learning and understanding of complex biomedical engineering concepts for university students. Integrated detailed 3D anatomical models, simulated laboratory environments, and interactive exercises to provide an immersive educational experience. The project aimed to bridge the gap between theoretical knowledge and practical application, resulting in improved student engagement and comprehension.

TeleHealth Pro

Coming Soon

(React, Node.js, Tailwind, Typescript, JavaScript.)

Designing and developing a telemedicine platform aimed at improving access to healthcare services for underserved and remote communities. Building secure patient-provider communication features, including appointment scheduling, virtual consultations, and digital health record management. Researching user experience considerations for patients with varying levels of digital literacy to ensure accessibility and ease of use. Exploring integration of AI-assisted symptom assessment and decision-support tools to enhance healthcare delivery. Applying biomedical engineering principles, software development, and healthcare workflow analysis to create a practical and scalable digital health solution.

ProjectFlow

Coming Soon

(React, Node.js, Typescript, Tailwind, JavaScript, Firebase)

ProjectFlow A project supervision and management platform designed to streamline MSc and undergraduate research workflows. The system helps students organize milestones, track progress, manage documentation, and maintain effective communication with supervisors through a centralized digital workspace.

CERTIFICATIONS

- AI Fluency: Frameworks and Foundations - Anthropic (2026)
- AI Agents Fundamentals - HuggingFace (2026)
- Essential Pain Management - Medical Aid International (2025)
- Ideation and Design Thinking - Center for Innovation and Technology Transfer (2025)
- Entrepreneurship and Leadership - Educate! (2019)
- Student Leader - Kitgum High School (2019)

LANGUAGES

- English (Fluent)
- Luo (Native)